National University of Computer and Emerging Sciences

Lahore Campus

Applied Machine Learning (AI4003)

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 3

Date: September 22nd 2026

Course Instructor(s)

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Attempt all the questions.

***CLO # 01: Demonstrate** the understanding of Supervised Machine learning algorithms.*

Q1: Demonstrate your understanding of Supervised Machine Learning by selecting the correct answer. [10 marks]

1. What does it mean for a linear regression model to generalize well?
   A. It memorizes the observations used during model training.
   B. It produces identical predictions for all observations in the training set.
   C. It is applicable only when the relationship between variables is nonlinear.
   D. It performs well when predicting outcomes for previously unseen observations.

2. Which ML technique is used in spam email detection?
   A. Clustering
   B. Reinforcement Learning
   C. Regression
   D. Classification

3. Which ML approach learns by interacting with an environment and receiving rewards?
   A. Supervised Learning
   B. Unsupervised Learning
   C. Reinforcement Learning
   D. Semi-supervised Learning

4. Which statement correctly describes the effect of regularization techniques in linear regression?
   A. Regularization is used to make a model more complex so that it fits the training data exactly.
   B. Ridge regression forces selected regression coefficients to become exactly zero.
   C. Lasso regression may reduce certain coefficients to exactly zero, allowing less important features to be excluded.
   D. Regularization can only be applied when a regression model contains one predictor.

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5. Which of the following factors can cause gradient descent to fail to converge to the global minimum in linear regression?
   A. Normalizing input features
   B. Using the mean squared error as the loss function
   C. Using a very large learning rate
   D. Using a very small learning rate

6. Which ML technique is used to avoid overfitting in neural networks?
   A. Dropout
   B. Gradient Descent
   C. Backpropagation
   D. Normalization

7. A logistic regression model is developed for a medical dataset in which only 8% of patients have a particular disease, while 92% do not. Which metric would be most appropriate for assessing the model's ability to correctly identify patients with the disease?
   A. Accuracy
   B. Precision
   C. Recall
   D. Mean Squared Error (MSE)

8. Which ML concept is related to bias-variance tradeoff?
   A. Overfitting and underfitting
   B. Feature scaling
   C. Normalization
   D. Hyperparameter tuning

9. What is the primary role of a loss function during the training of a machine learning model?
   A. It determines the architecture and number of layers in the model.
   B. It defines the objective that the optimization algorithm seeks to minimize during training.
   C. It is used only to evaluate the model after training is completed.
   D. It has no significant effect on how the model learns from the training data.

10. A logistic regression model is used to classify 10 patients as either positive or negative. The model produces the following results:

	Actual Positive	Actual Negative
Predicted Positive	4	2
Predicted Negative	1	3

What is the recall of the logistic regression model?
   A. 60%
   B. 70%
   C. 80%
   D. 90%

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*CLO # 02: **Design** and implement a supervised machine learning model to solve real-world problems and evaluate its performance using appropriate metrics*

Q2: Design a Supervised Machine Learning model for the given problem. [10 marks]

A. Given the following dataset, (1, 1.3), (4, 3.75), (5, 2.25), consider the linear regression model with ridge regression. Assume $\lambda = 5$.

$$\hat{y} = \theta_0 + \theta_1 x$$

Using batch gradient descent, perform one iteration (epoch) starting from $\theta_0 = 0$ and $\theta_1 = 1$. Assume the loss function is

$$SSE = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + Ridge\ penalty$$

And the learning rate = 0.01. Determine the updated values of θ_0 and θ_1

Note: Do not use normalization.

*CLO # 02: **Design** and implement a supervised machine learning model to solve real-world problems and evaluate its performance using appropriate metrics*

Q3: Design a Supervised Machine Learning model for the given problem. [3+3+4 marks]

A logistic regression model for predicting whether a student will pass (1) or fail (0) an exam is given by $z = -4 + 0.8x$ where x represents the number of hours studied.

The predicted probability of passing is calculated using the sigmoid function:

$$P(Y = 1|x) = \frac{1}{1 + e^{-z}}$$

The classification threshold is 0.5.

Answer the following questions:

A. Calculate the probability that a student who studies 6 hours will pass. Based on the threshold of 0.5, classify the student as Pass or Fail.
B. Calculate the probability of passing for a student who studies 4 hours. Classify the 4-hour student as Pass or Fail.
C. Determine the minimum number of hours a student must study for the model to predict Pass, using the classification threshold of 0.5.

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